WASP-35b

R-1519

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-35_181219` · created 2026-08-01T16:50:58+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458471.7389 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1474 +/- 0.0065
Transit depth (Rp/Rs)^2	2.17 +/- 0.19 [%]
Orbital Inclination (inc)	88.17 +/- 0.77
Airmass coefficient 1 (a1)	1.184 +/- 0.01
Airmass coefficient 2 (a2)	-0.1206 +/- 0.006
Scatter in the residuals of the lightcurve fit is	0.88 %
Best Comparison Star	None
Optimal Aperture	5.01
Optimal Annulus	13.6
Transit Duration (day)	0.1347 +/- 0.0033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1474 ± 0.0065 vs 0.1241 (deviation 2.93σ)
Duration fit vs published	0.1347 d vs 0.1317 d (deviation 0.75σ)
Mid-time O–C	0.5 min
Depth SNR	18.5
Residual scatter	0.88 %

Light curve

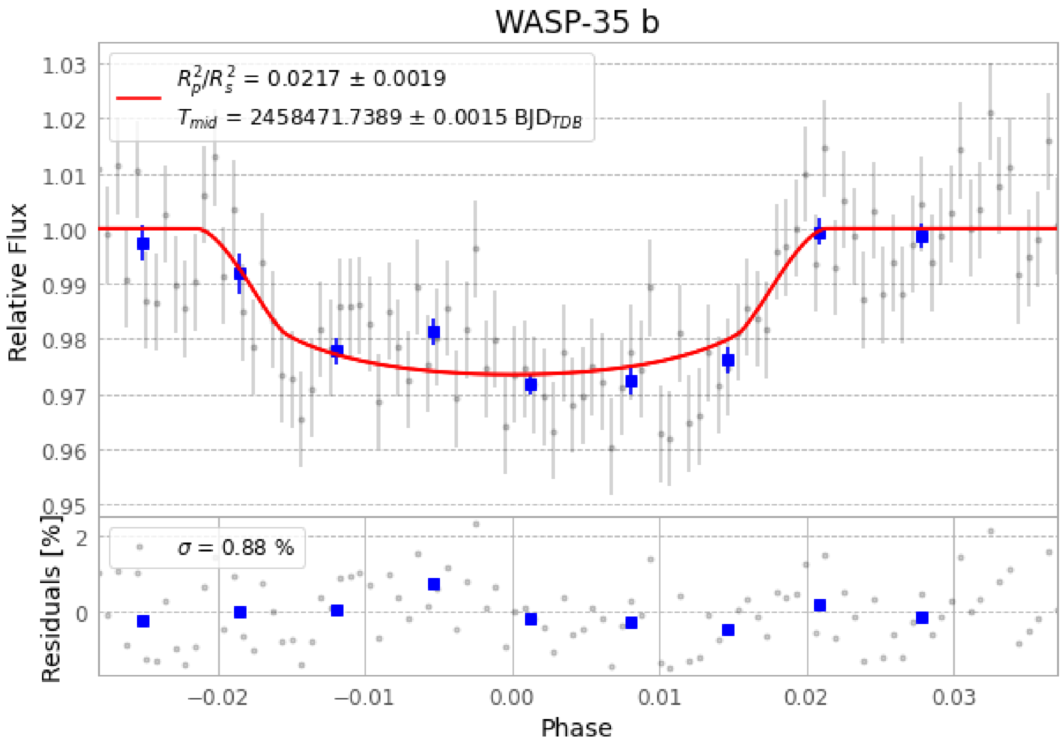

Figure 1 — FinalLightCurve_WASP-35 b_2018-12-18.png (SHA-256 9cf150eb11158160…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [292, 203], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 e2ed7a463c7530b1…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

100 FITS frames (66 MB), `WASP-35181219032712.FITS` → `WASP-35181219082412.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-181219.log (b2251f113abc…), FinalLightCurve_WASP-35 b_2018-12-18.png (9cf150eb1115…), FinalLightCurve_WASP-35 b_2018-12-18.csv (16fa688fc8ff…)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)